Data Types

…and the art of visualization

Data types

Numerical (measures)

Categorical (dimensions)

1, 2, 3… vs A, B, C…

Numerical (*measures*)

Quantitative data attribute

How many, how long? ...

Can be calculated and/or aggregated to a single number (> metrics):

Number of occurrences

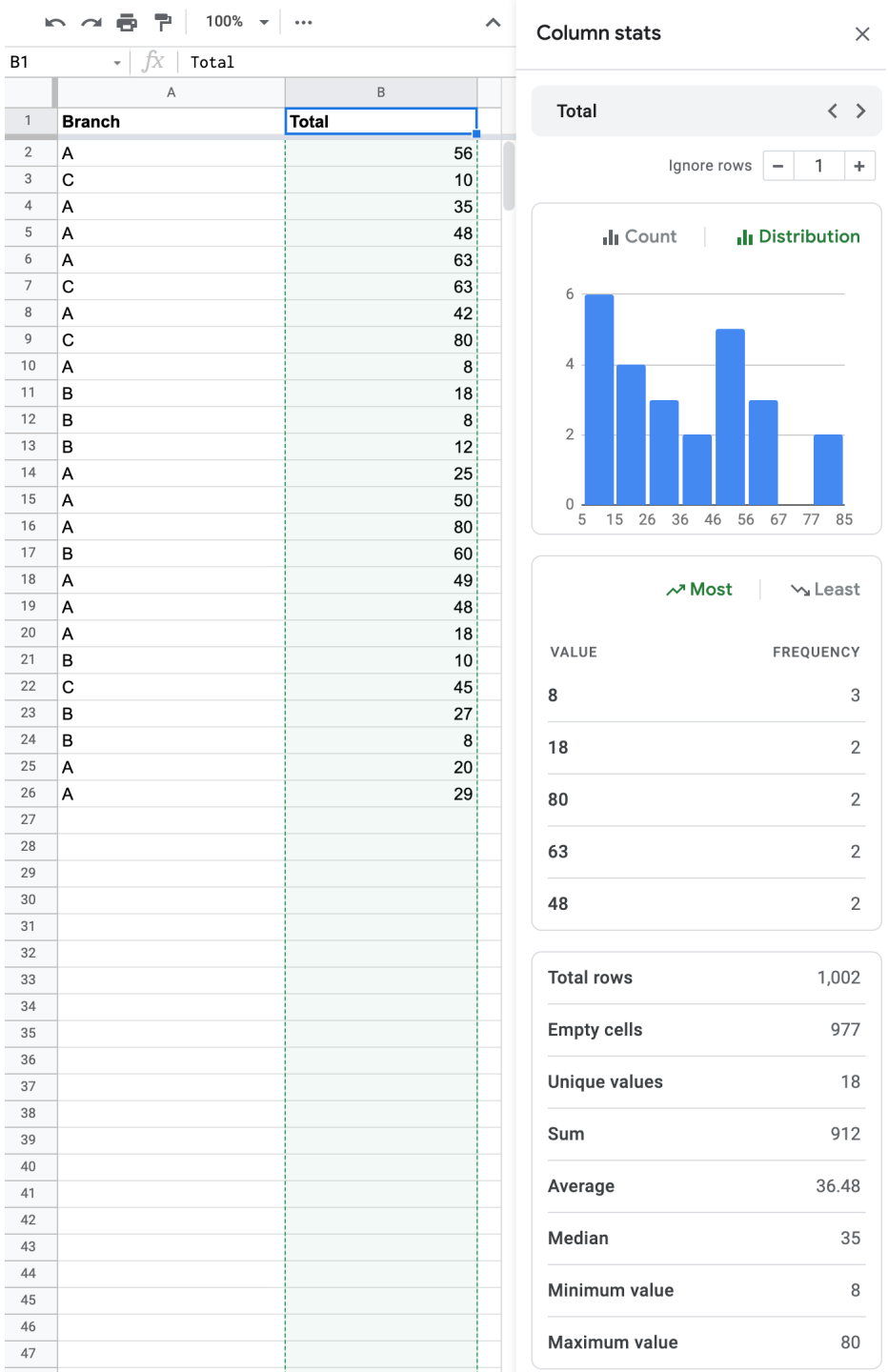
Total

Mean, median, mod

Standard deviation

Min, Max

...



	A	B
1	Branch	Total
2	A	56
3	C	10
4	A	35
5	A	48
6	A	63
7	C	63
8	A	42
9	C	80
10	A	8
11	B	18
12	B	8
13	B	12
14	A	25
15	A	50
16	A	80
17	B	60
18	A	49
19	A	48
20	A	18
21	B	10
22	C	45
23	B	27
24	B	8
25	A	20
26	A	29

Numerical (*measures*)

Can be broken down by dimensions

Examples:

Product price

Duration

Number of transactions

Length, width, volume

Weight

...

Month	*Week*	SUM of Total
− 1	1	104
	2	8
	3	60
	4	18
	5	48
− 2	6	121
	7	25
	8	38
	9	167
− 3	9	27
	10	84
	11	67
	13	143

Categorical (*dimensions*)

Qualitative (descriptive) data attribute

Categorical data

Who. What? Where? When.

It cannot be calculated. Nor can metrics (mean, sum, etc.) be calculated for it.

100%

A1 | Branch

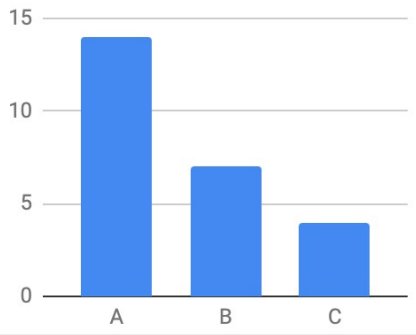
	A	B
1	Branch	Total
2	A	56
3	C	10
4	A	35
5	A	48
6	A	63
7	C	63
8	A	42
9	C	80
10	A	8
11	B	18
12	B	8
13	B	12
14	A	25
15	A	50
16	A	80
17	B	60
18	A	49
19	A	48
20	A	18
21	B	10
22	C	45
23	B	27
24	B	8
25	A	20
26	A	29
27		
28		
29		
30		
31		
32		
33		
34		

Column stats

Branch

Ignore rows − 1 +

Count | Distribution



Most | Least

VALUE	FREQUENCY
A	14
B	7
C	4

Total rows	1,002
Empty cells	977
Unique values	3

All stats include data in hidden or filtered rows

Categorical (*dimensions*)

It helps to understand and organize quantitative measures

- **Describes them**
- **Provides context**
- **Determines the level of detail in reports and analyses**

It can be used for:

- **Categorization**
- **Segmentation**
- **Dividing another measure into several different groups**

Examples:

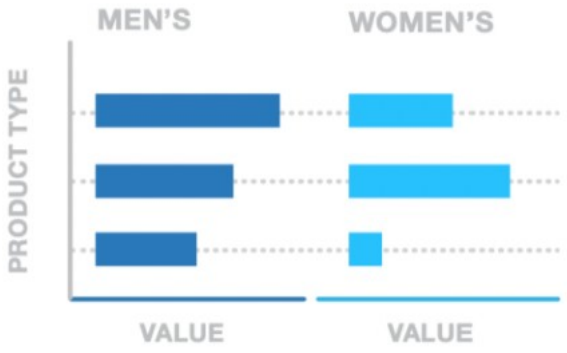
Sales by region > subregions

Sales by year > quarter > month > week

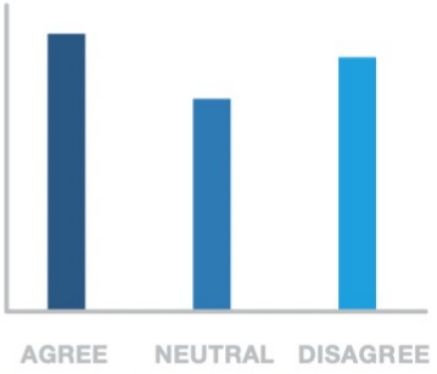
Month	*Week*	SUM of Total
− 1	1	104
	2	8
	3	60
	4	18
	5	48
− 2	6	121
	7	25
	8	38
	9	167
− 3	9	27
	10	84
	11	67
	13	143

Dimension

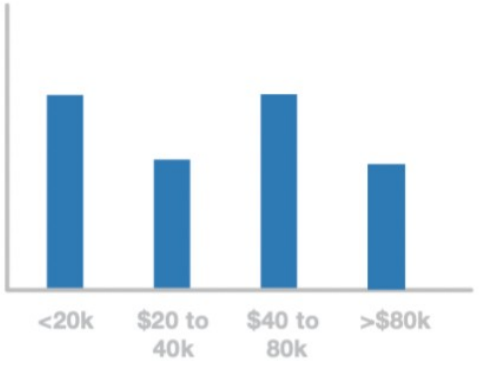
Nominal dimensions: these are usually names, labels, etc.



Order dimensions: their elements can be ordered (sequenced)



Interval dimensions: their values represent intervals of numeric values



data-to-viz.com

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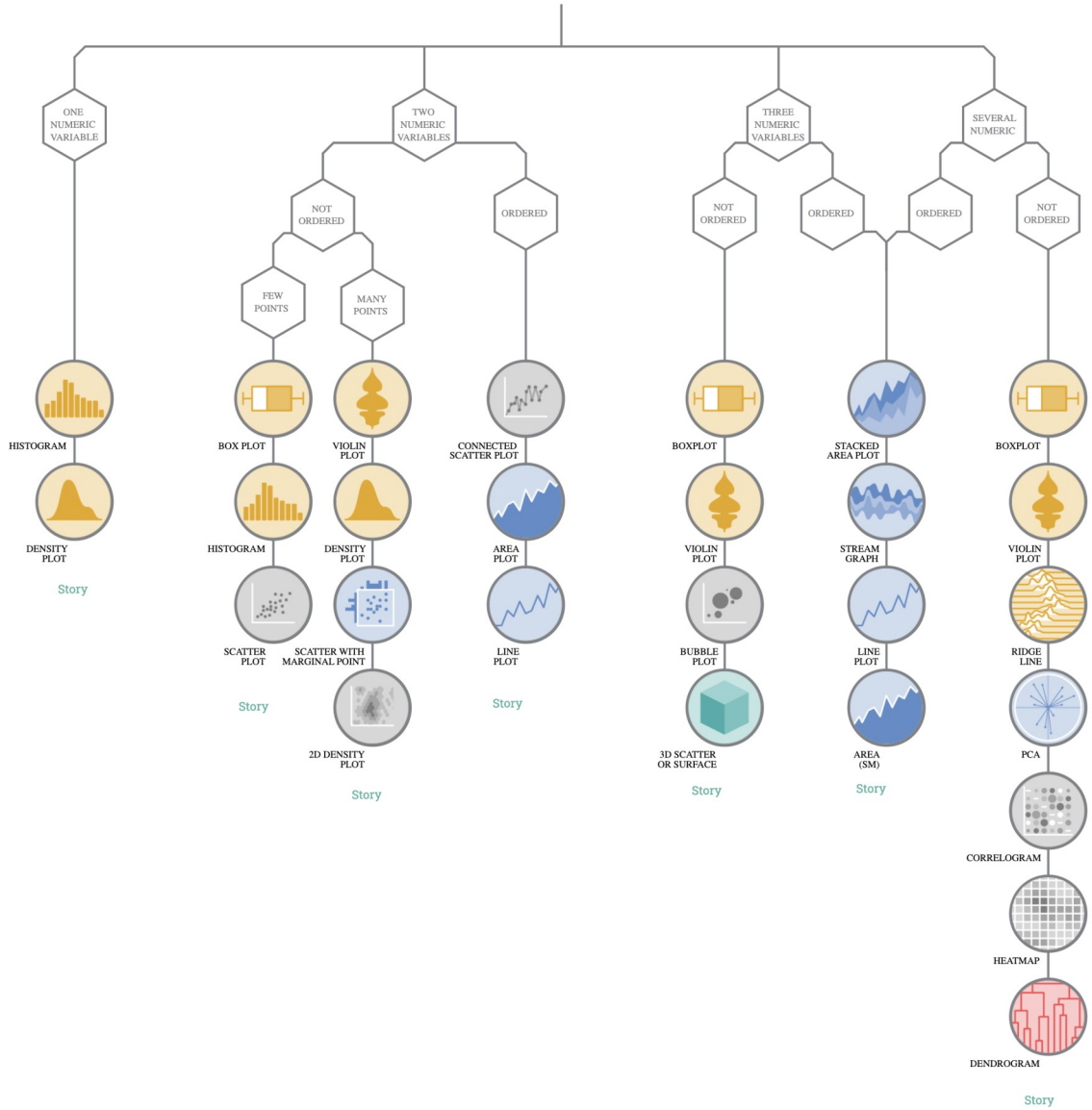
What kind of data do you have? Pick the main type using the buttons below. Then let the decision tree guide you toward your graphic possibilities.

Numeric | Categoric | Num & Cat | Maps | Network | Time series



sap.com/documents/2022/01/5e49ac93-127e-0010-bca6-c68f7e60039b.html

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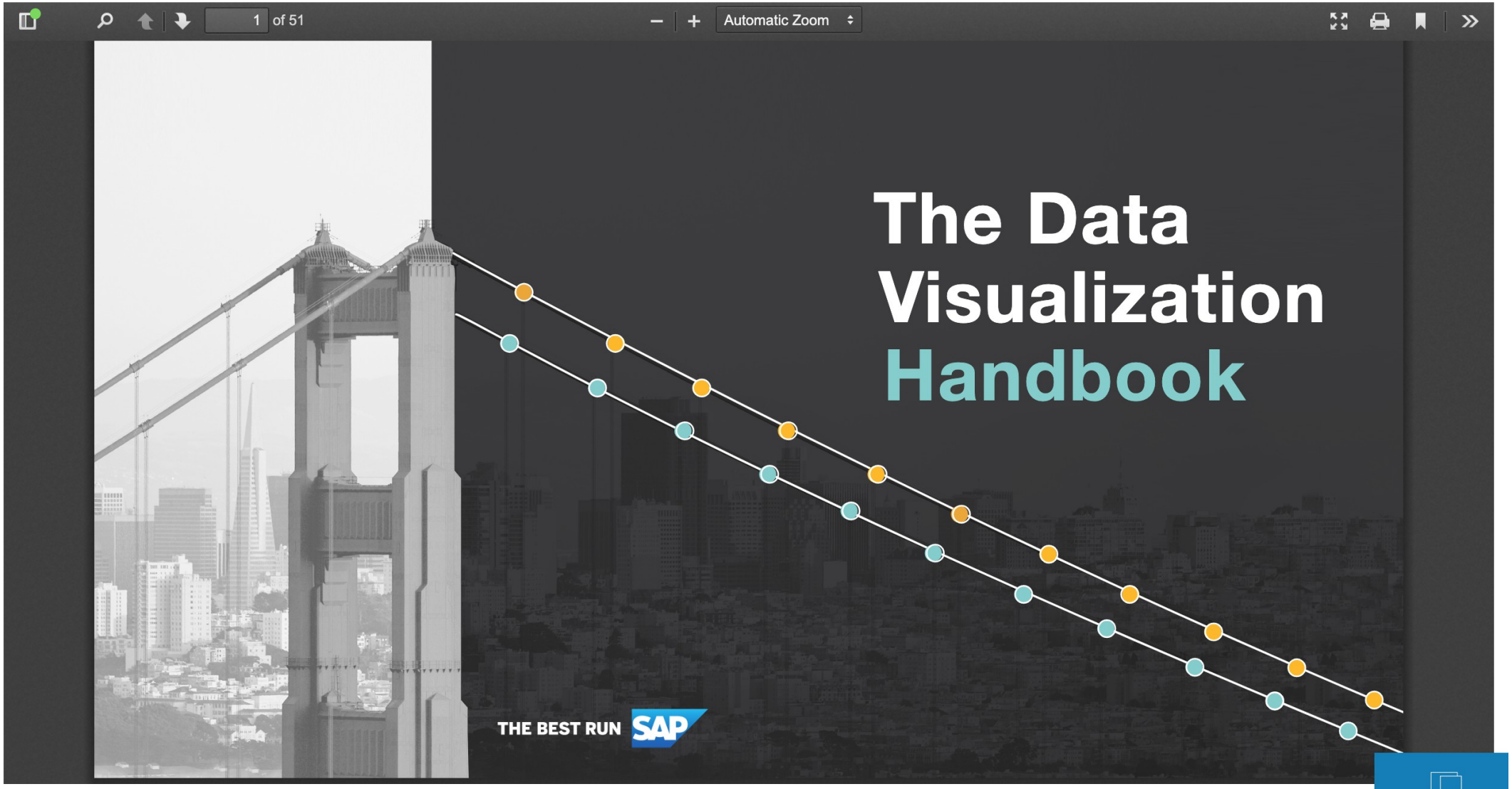

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SAP Analytics Cloud - Data Visualization Handbook

With this handbook, you will discover how to approach your data, howto make the most of elements within data visualizations, and decide which visualizationstell your audience the best story.

Download the Document

sap.com/documents/2022/01/5e49ac93-127e-0010-bca6-c68f7e60039b.html

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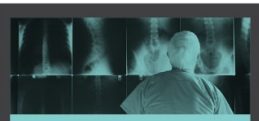
Selecting the Right Visualizations

Change Over Time

Shows how a measure changes over time, and allows the user to highlight temporal trends

- Line and Area Charts: Highlights potential trends in data
- Column Line Chart: compares two measures over a period of time
- Bullet: Indicates progress towards a goal
- Time Series Chart: Shows trends over time



Comparison

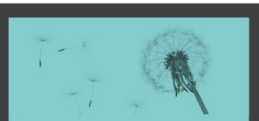
Shows the comparison of categorical values, where the data does not have any intrinsic order, for example, a list of products

- Bar Chart: Used for comparing categorical values
- Waterfall chart: Highlights the cumulative effect of sequential data
- Trellis: Shows partitions of a dataset across multiple views
- Merimekko: Communicates through width

Ranking

Shows the top or bottom N values to emphasize the largest, or smallest values

- Bar Charts: Shows categorical values in decreasing or increasing order
- Cluster Bubble: Shows the value of a measure based on two dimensions



Part-To-Whole

Shows how the categories contribute to the whole value

- Bar Chart: Set to % scale
- Stacked Bar Chart: Shows overall measure total
- Pie Chart: Compares percentage values

Distribution

Shows how a measure is spread across its domain

- Box Plot: Shows distributions for different categorical values
- Heat and Tree Map: Shows the distribution of measure values
- Radar: Sees the distribution of values across dimensions

Correlation

Shows, whether there is a potential correlation between two measures

- Scatter Plot: Highlights potential correlation of two measures
- Bubble Chart: Displays values that consist of three variables

Geographical Information and Maps

Shows the geographical distribution of measure values

- Choropleth Chart: Highlights geographical data by colouring geographical areas according to their measure values
- Geo Bubble Chart: Highlights geographical data by showing them as bubbles on a map

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